Machine Learning Project

Customer Segmentation using K-Means Clustering

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Problem Statement

Businesses often treat all customers the same, lacking personalized marketing strategies. Understanding customer behavior becomes difficult without proper segmentation. "We need to group customers based on behavior."



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Objective

Our project aims to segment customers using machine learning techniques, specifically K-Means clustering. We seek to identify meaningful behavior patterns within customer data and ultimately improve business decision making through data-driven insights.



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Tech Stack Tools & Libraries

Our customer segmentation solution is built using industry-standard tools:

- Python as the core programming language, Streamlit for interactive web deployment.
- Scikit-learn for machine learning algorithms, Pandas for data manipulation.
- Plotly for dynamic visualizations.

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ML Pipeline Workflow

Our customer segmentation process follows a streamlined workflow:

- Upload data
- Preprocess & clean
- Feature Selection
- Train K-Means model
- Predict customer segments.

Each step builds upon the previous to deliver accurate clustering results.



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What is K-Means Clustering?

K-Means is an unsupervised machine learning algorithm that groups similar data points into clusters. It works by calculating distances between points and iteratively assigning them to the nearest cluster center, making it ideal for customer segmentation.

For Optimization of our K - Means Model we use :

- Elbow Method (Inertia)
- Silhouette Score

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Results and Customer Segments

Customer clusters successfully formed using K-Means algorithm with PCA visualization. Identified segments:

- High Income - High Spending
- High Income - Low Spending
- Low Income - High Spending
- Low Income - Low Spending

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Conclusion

We successfully built a complete ML pipeline for customer segmentation using K-Means clustering. This solution enables targeted marketing strategies and personalized customer engagement, making it directly applicable to real-world business scenarios.



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